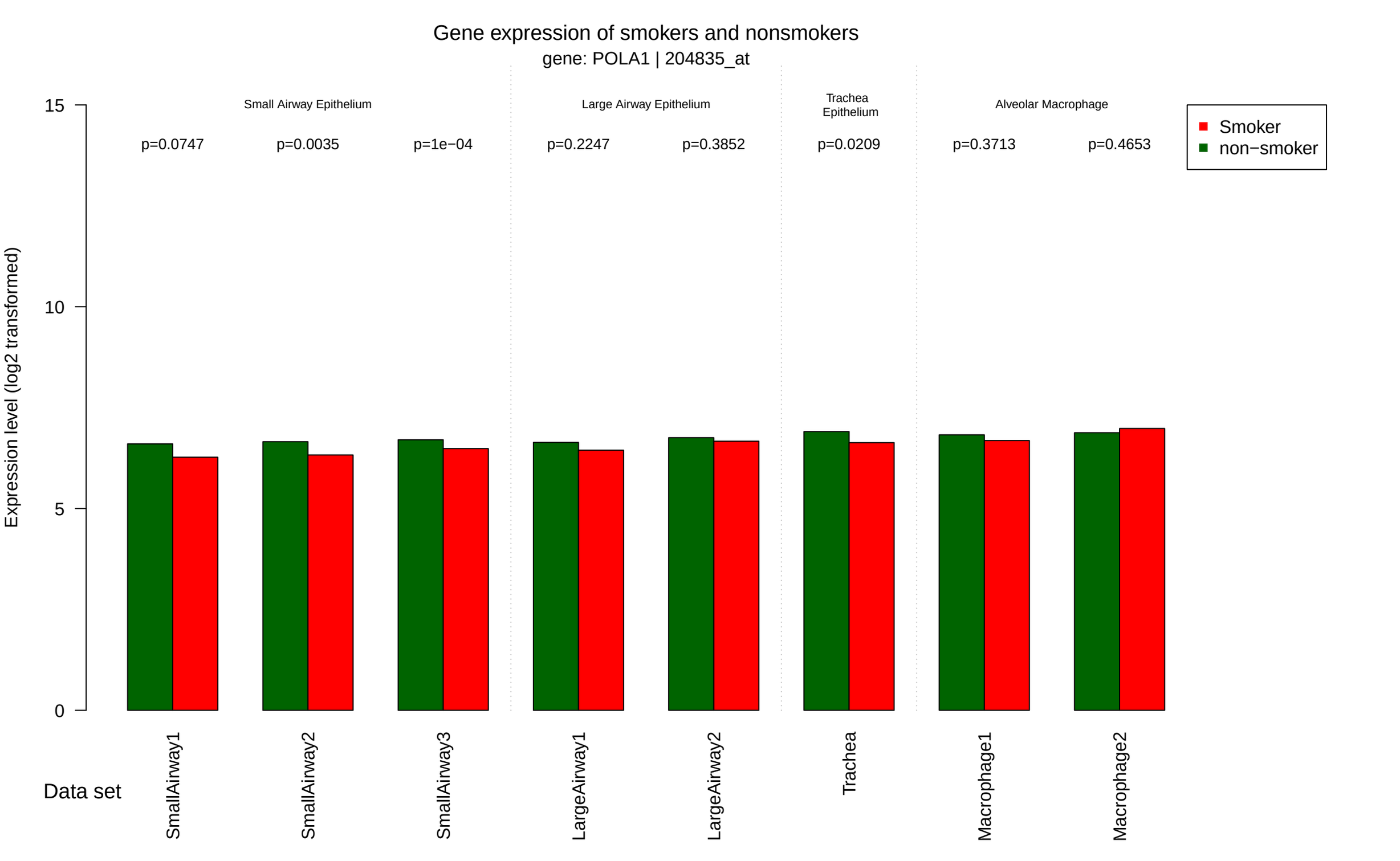
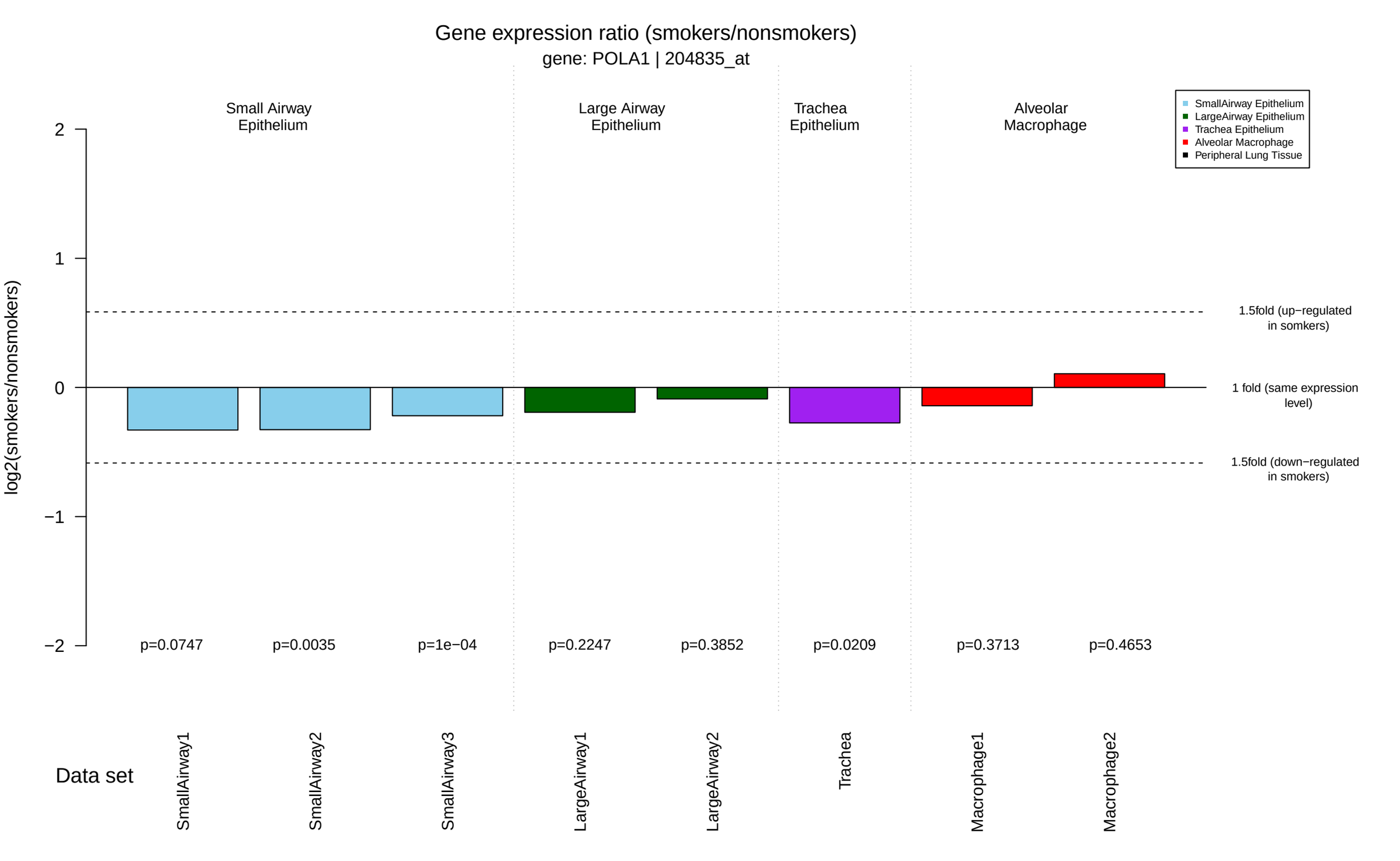
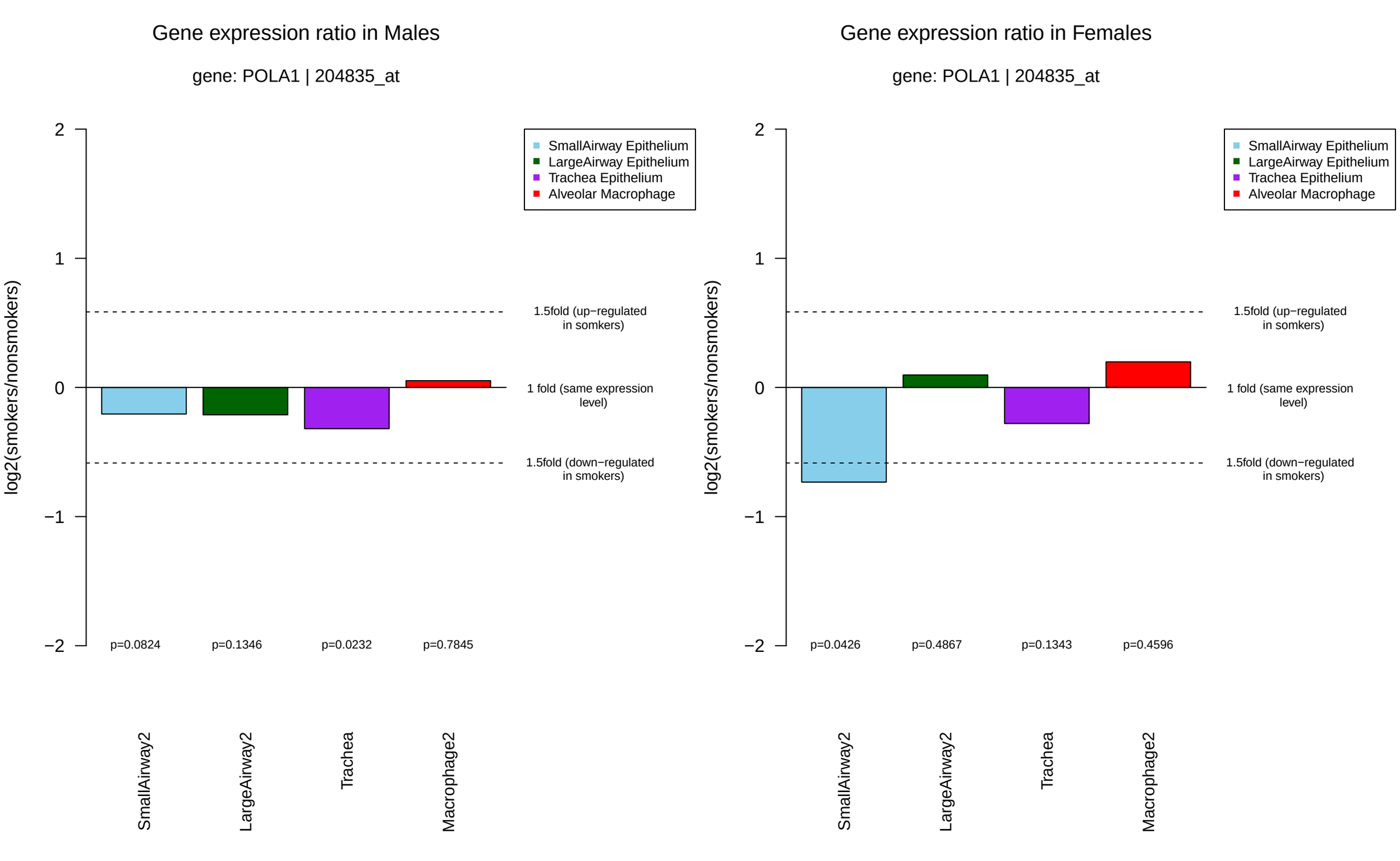
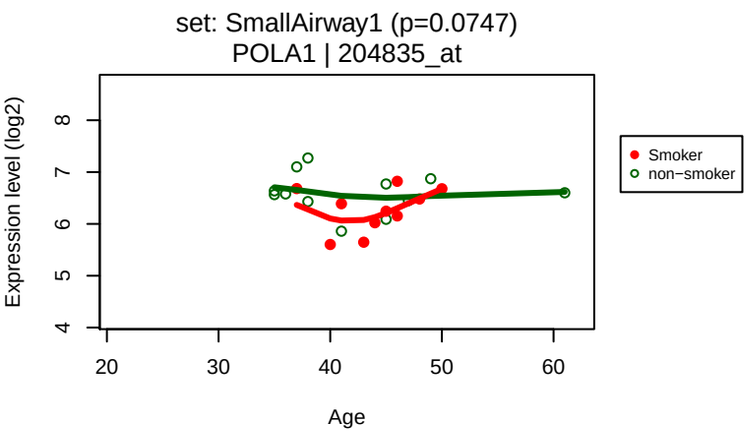
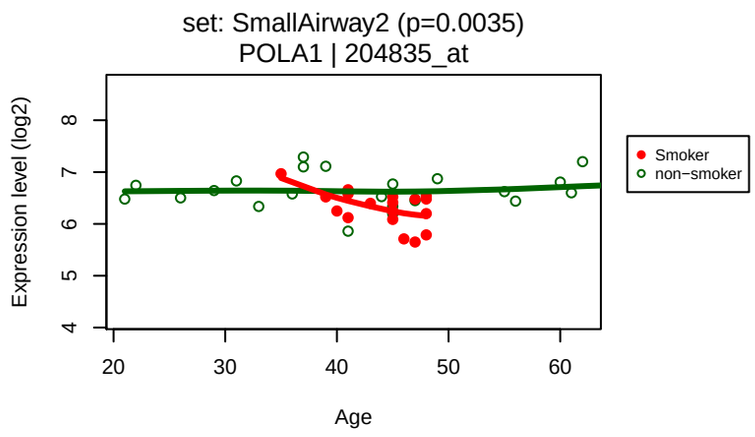
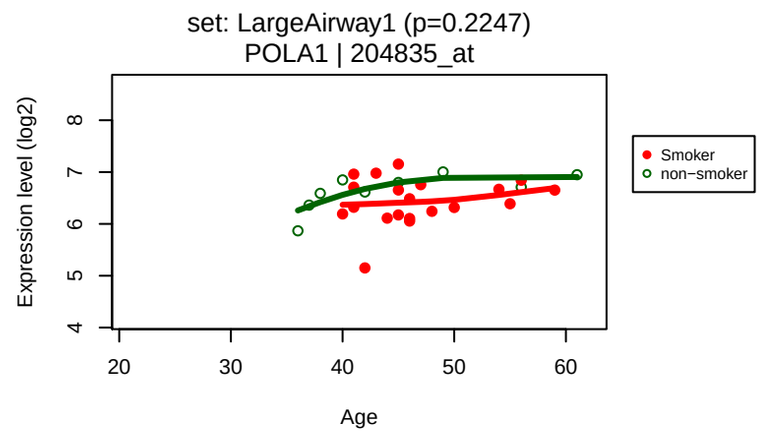

Gene expression of smokers and nonsmokers
gene: POLA1 | 204835_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0747
p=0.0035
p=1e−04
p=0.2247
p=0.3852
p=0.0209
p=0.3713
p=0.4653
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

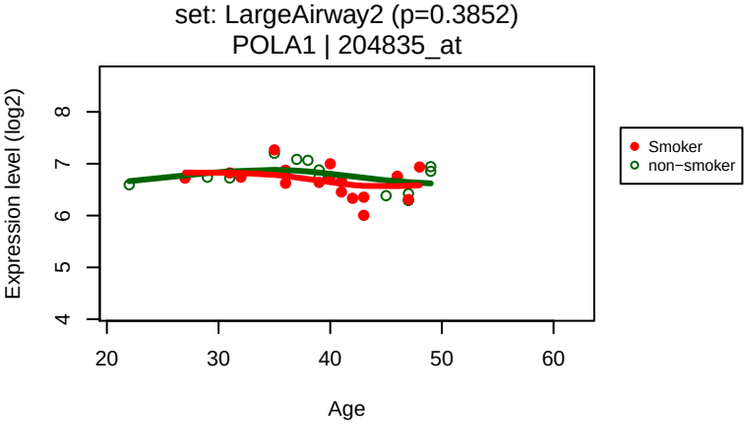

Gene expression ratio (smokers/nonsmokers)
gene: POLA1 | 204835_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0747
p=0.0035
p=1e−04
p=0.2247
p=0.3852
p=0.0209
p=0.3713
p=0.4653
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

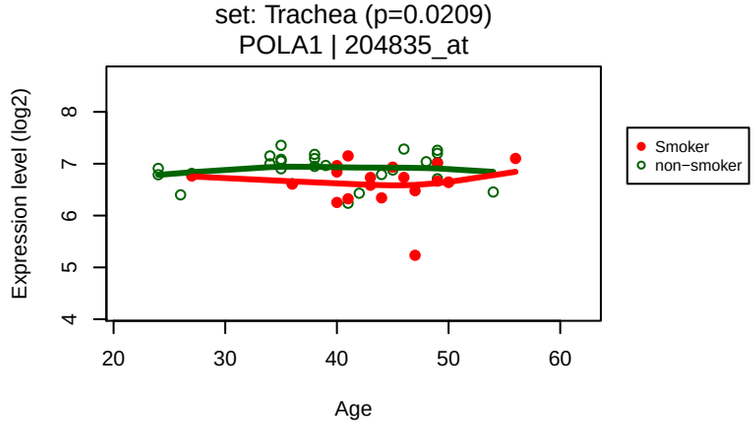

Gene expression ratio in Males
gene: POLA1 | 204835_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0824
p=0.1346
p=0.0232
p=0.7845
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLA1 | 204835_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0426
p=0.4867
p=0.1343
p=0.4596
SmallAirway2
LargeAirway2
Trachea
Macrophage2

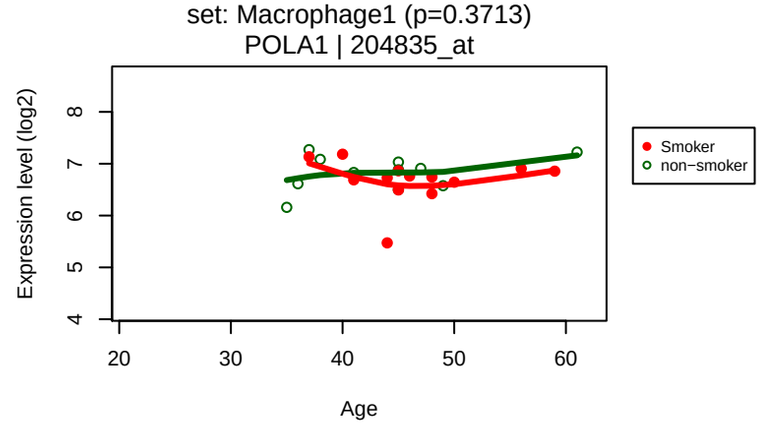

set: SmallAirway1 (p=0.0747)
POLA1 | 204835_at
Expression level (log2)
Age
Smoker
non−smoker

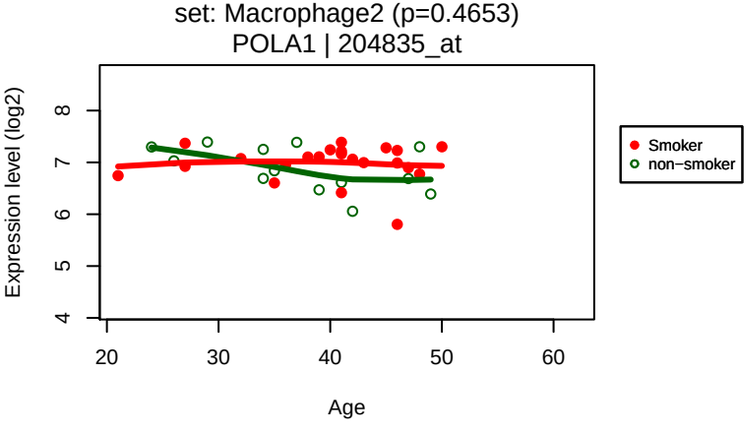

set: SmallAirway2 (p=0.0035)
POLA1 | 204835_at
Expression level (log2)
Age
Smoker
non−smoker

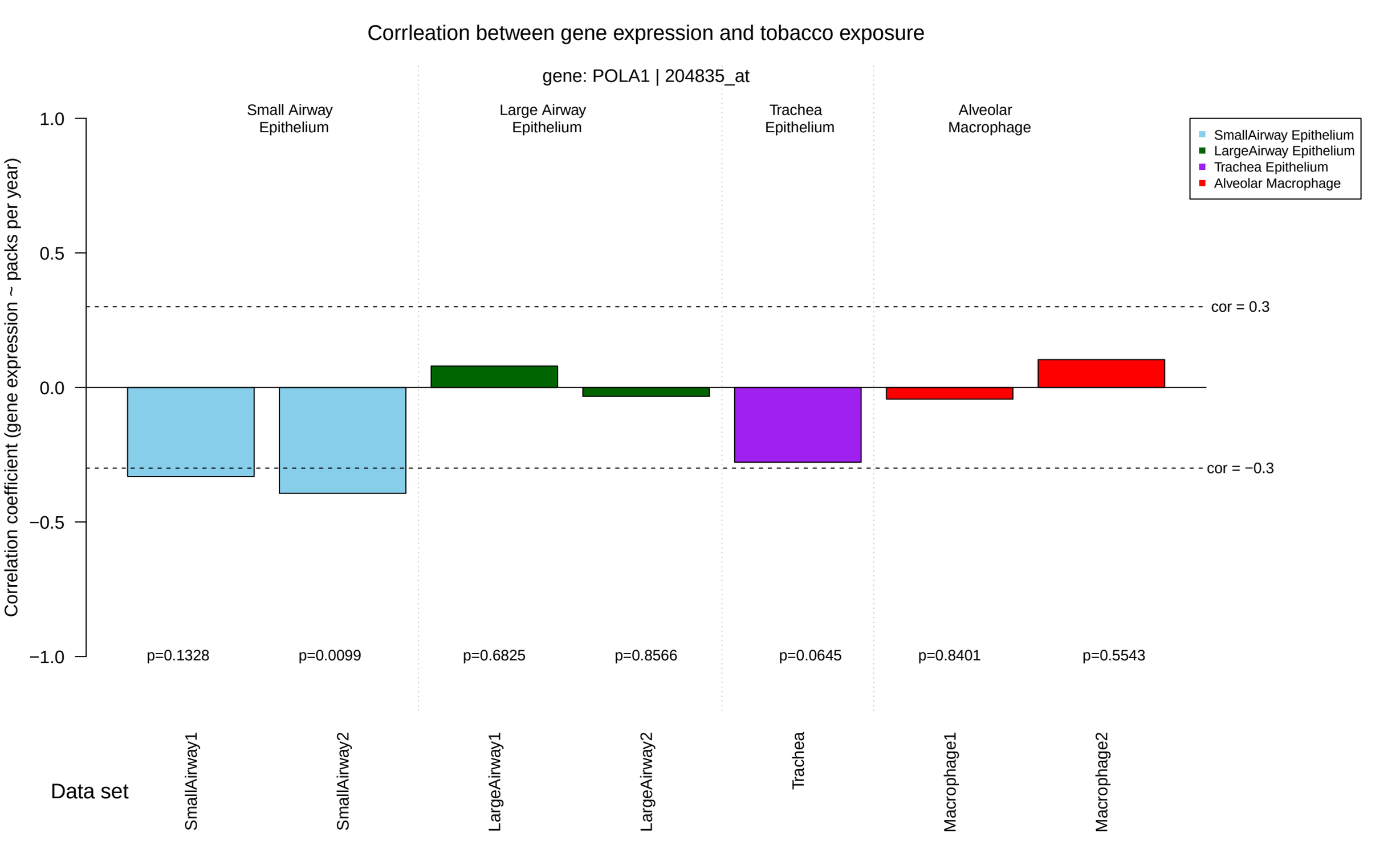

set: LargeAirway1 (p=0.2247)
POLA1 | 204835_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3852)
POLA1 | 204835_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0209)
POLA1 | 204835_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3713)
POLA1 | 204835_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4653)
POLA1 | 204835_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLA1 | 204835_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1328
p=0.0099
p=0.6825
p=0.8566
p=0.0645
p=0.8401
p=0.5543
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2